

Arnav Surve

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EDUCATION

California State University - Sacramento

Bachelor of Science, Computer Science

Sacramento, CA

Aug. 2024 – May 2026

Purdue University

Bachelor of Science, Cybersecurity – Dean's List, Fall 2023

West Lafayette, IN

Aug. 2022 – May 2024

SKILLS

Languages: Go, Python, TypeScript, JavaScript, SQL, PostgreSQL, Java, HTML/CSS, YAML, JSON, Bash
Frameworks: React, Node.js, FastAPI, Express.js, Echo, GORM, GraphQL, Prisma, Model Context Protocol
Cloud/DevOps: AWS, Lambda, Glue, RDS, ECR, EKS, Kubernetes, Docker, Terraform, Git, GitHub Actions
Data/Blockchain: PostgreSQL, Redis, ETL, VSAM, Blockchain, Smart Contracts, Tezos, NFTs, Web3
Practices: CI/CD, Microservices, REST APIs, Agile/Scrum, DevOps, Monitoring, IaC

EXPERIENCE

Software Engineer Intern

California Department of Motor Vehicles

March 2025 – Present

Sacramento, CA

- Architected and deployed end-to-end ETL pipeline processing mainframe VSAM dataset exports for vehicle registrations and driver licenses, successfully migrating 197 million records to production RDS PostgreSQL database serving California Highway Patrol, FBI, and other federal law enforcement agencies.
- Developed comprehensive data processing solution using AWS Lambda and Glue for initial bulk load and daily synchronization pipelines handling 200-300k daily records with upsert/delete operations.
- Implemented entire infrastructure as code using Terraform, managing deployment lifecycle across dev, UAT, and production environments while supporting comprehensive UAT testing throughout development cycle.
- Refactored and deployed distributed vehicle title management microservices blockchain system using TypeScript monorepo with 5 interconnected Tezos smart contracts (FA2 NFTs), implementing multitenant role-based access control and chain bridge worker pools for database-blockchain synchronization.
- Deployed self-hosted private Tezos blockchain node infrastructure on Kubernetes using ECR and EKS, orchestrating containerized microservices with production-ready CI/CD pipelines and comprehensive monitoring.

IT Infrastructure Intern

MITER Brands – Milgard Manufacturing

June 2024 – Aug. 2024

Tacoma, WA

- Implemented APM pipeline optimizations in Site24x7 and New Relic, reducing alert redundancy by 60% across 6 teams and 2 business units.
- Developed REST API endpoint monitors identifying critical discrepancies, resulting in an average 20% latency reduction due to increased observability.
- Automated Active Directory compliance audits using PowerShell scripting, improving operational efficiency by 20%.

Web Developer

Purdue University Office of Engagement

Aug. 2022 – May 2023

West Lafayette, IN

- Delivered full-stack research grant application platform using Agile Scrum methodology in 10-person development team.
- Built RESTful API with Express.js and PostgreSQL using Prisma ORM, implementing federated SSO authentication.
- Containerized application with Docker for deployment to Purdue University server cluster infrastructure.

PROJECTS

Grace Programming Language | Go, Compiler Design, Type Systems, Code Generation

2024

- Built statically-typed programming language (8,407 LOC) that compiles to readable COBOL, featuring complete compiler toolchain with lexer, recursive descent parser using Pratt parsing, and comprehensive AST for semantic analysis.
- Designed novel width-aware type system with compile-time COBOL PIC clause validation, automatic width inference for arithmetic expressions, and overflow detection for mainframe compatibility.
- Implemented sophisticated code generation translating modern constructs (functions, structs, type inference) into structured COBOL with automatic temporary variable management.

Grace Infrastructure Platform (graceinfra.org) | Go, Zowe CLI, DAG Execution, Mainframe Modernization

2024

- Engineered hybrid mainframe-cloud orchestration platform (8,366 LOC) modernizing COBOL workflows through declarative YAML configuration and Infrastructure-as-Code principles.
- Built sophisticated DAG execution engine with concurrent job processing, dependency resolution, and intelligent state management for complex multi-step mainframe-cloud workflows.
- Implemented extensible plugin architecture supporting multiple job handlers (compile, linkedit, execute, shell) with automatic cleanup of temporary mainframe datasets and comprehensive error handling.